Math 104 Crash Course

Midterm 2 · Lectures 9–14 · Spring 2026

0 Contents

1 Series and Convergence

Big Picture

A **series** $\sum a_n$ is just a sequence in disguise: the sequence of partial sums $s_N = \sum_{n=1}^{N} a_n$. All your sequence machinery applies directly. The hard part is that you usually can't compute s_N in closed form, so you need indirect tests.

1.1 Cauchy Criterion

A sequence $\{s_n\}$ converges if and only if it is *Cauchy*: for every $\varepsilon > 0$, there exists N such that $m, n \geq N \Rightarrow |s_m - s_n| < \varepsilon$.

For a series, $|s_m - s_n| = \left|\sum_{k=n+1}^{m} a_k\right|$. So a series is Cauchy iff its tails can be made arbitrarily small.

Corollary 1.1. *If $\sum a_n$ converges, then $a_n \to 0$.* ***Converse fails****: the harmonic series $\sum 1/n$ diverges even though $1/n \to 0$.*

Key Insight

The n-th term test only rules out convergence. If $a_n \not\to 0$, you're done—it diverges. But $a_n \to 0$ tells you nothing; you need a real test.

1.2 Monotone Sequences and Finding Limits

Theorem 1.2. *A monotone bounded sequence converges.*

How to find the limit when you have a recurrence $x_{n+1} = f(x_n)$:

1. Show the sequence is monotone and bounded (so it converges to some L).
2. Take $n \to \infty$ in the recurrence: $L = f(L)$ (valid because f is continuous).
3. Solve for L.

1.3 Comparison and Condensation Tests

Theorem 1.3 (Comparison Test)**.** *If $|a_n| \leq c_n$ for all large n and $\sum c_n$ converges, then $\sum a_n$ converges.*

If $a_n \geq d_n \geq 0$ and $\sum d_n$ diverges, then $\sum a_n$ diverges.

Theorem 1.4 (Cauchy Condensation Test)**.** *Let $\{a_n\}$ be non-negative and decreasing. Then*

$$\sum_{n=1}^{\infty} a_n \quad \textit{converges} \quad \Longleftrightarrow \quad \sum_{k=0}^{\infty} 2^k a_{2^k} \quad \textit{converges.}$$

Key Insight

Why does condensation work? The idea is that on each dyadic block $[2^k, 2^{k+1})$, the decreasing sequence a_n is sandwiched between a_{2^k} (the biggest) and $a_{2^{k+1}}$ (the smallest). There are 2^k terms in the block, so the block sum is sandwiched between $2^k a_{2^{k+1}}$ and $2^k a_{2^k}$. The condensed series $\sum 2^k a_{2^k}$ captures exactly this block size.
Application to p-series: $a_n = 1/n^s$ gives $2^k a_{2^k} = 2^k/2^{ks} = 2^{k(1-s)}$, which is a geometric series converging iff $2^{1-s} < 1$ iff $s > 1$.

1.4 Root Test and Ratio Test

Theorem 1.5 (Root Test)**.** *Let $\alpha = \limsup |a_n|^{1/n}$.*

- $\alpha < 1$*: $\sum a_n$ converges absolutely.*
- $\alpha > 1$*: $\sum a_n$ diverges.*
- $\alpha = 1$*: inconclusive.*

Theorem 1.6 (Ratio Test)**.** *Let $\beta = \limsup |a_{n+1}/a_n|$ and $\alpha_* = \liminf |a_{n+1}/a_n|$.*

- $\beta < 1$*: converges absolutely.*
- $\alpha_* > 1$*: diverges.*
- $\beta = 1$ *or* $\alpha_* = 1$*: inconclusive.*

Key Insight

Root test is strictly stronger than the ratio test. If the ratio test gives a limit L, then the root test also gives L (because $\liminf |a_{n+1}/a_n| \le \limsup |a_n|^{1/n} \le \limsup |a_{n+1}/a_n|$). But the root test can succeed where the ratio test fails (e.g., $\limsup = 1$ for ratio but < 1 for root).
When to use ratio: factorials ($n!$) and pure exponentials (c^n), where ratios simplify cleanly.
When to use root: n-th powers $(c_n)^n$, since the $\limsup^{1/n}$ kills the power.

1.5 Power Series and Radius of Convergence

Definition 1.7. A **power series** centered at a is $\sum_{n=0}^{\infty} c_n (x-a)^n$. Its **radius of convergence** is

$$R = \frac{1}{\limsup_{n\to\infty} |c_n|^{1/n}}$$

with conventions $1/0 = +\infty$ and $1/\infty = 0$.

Theorem 1.8 (Cauchy–Hadamard)**.** *The series converges absolutely for $|x-a| < R$ and diverges for $|x-a| > R$. At $|x-a| = R$, check separately.*

Key Insight

The formula for R comes directly from the root test: $\limsup |c_n(x-a)^n|^{1/n} = |x-a| \cdot \limsup |c_n|^{1/n}$. This is < 1 iff $|x-a| < 1/\limsup |c_n|^{1/n} = R$.
Key examples:

- $e^x = \sum x^n/n!$: $|c_n|^{1/n} = (1/n!)^{1/n} \to 0$, so $R = \infty$.
- $\sum n! x^n$: $|c_n|^{1/n} = (n!)^{1/n} \to \infty$, so $R = 0$ (converges only at $x = 0$).
- $\sum x^n$: geometric, $R = 1$. Check endpoints: $x = \pm 1$ both diverge.

2 Differentiation

Big Picture

Differentiation is about local linear approximation. The derivative $f'(x)$ is the slope of the best linear fit to f near x. The MVT is the bridge between pointwise derivative information and global behavior—it's the most-used theorem in the course.

2.1 The Derivative

Definition 2.1. For $f : [a, b] \to \mathbb{R}$ and $x \in [a, b]$, the **derivative** is

$$f'(x) = \lim_{t \to x} \frac{f(t) - f(x)}{t - x},$$

provided this limit exists.

Theorem 2.2 ($C^1 \subset C^0$)**.** *Differentiable at x implies continuous at x. Converse fails: $f(x) = |x|$ is continuous at 0 but not differentiable.*

Proof sketch. Write $f(t) - f(x) = \frac{f(t)-f(x)}{t-x} \cdot (t - x)$. As $t \to x$: the first factor $\to f'(x)$ and the second $\to 0$. Product $\to 0$. □

2.2 Simple Discontinuities

Definition 2.3. A function f has a **simple discontinuity** at x if both one-sided limits $f(x^+)$ and $f(x^-)$ exist, but either:

- $f(x^+) \neq f(x^-)$ (a **jump discontinuity**), or
- $f(x^+) = f(x^-)$ but $\neq f(x)$ (a **removable discontinuity**).

If one-sided limits don't exist, it's a **discontinuity of the second kind** (e.g., $\sin(1/x)$ at 0).

Key Insight

Simple discontinuities are "well-behaved" in the sense that the one-sided limits exist. This matters for integration: a function with only finitely many simple discontinuities is still Riemann integrable (its discontinuities form a finite, hence zero, set).

2.3 Mean Value Theorem

Theorem 2.4 (Rolle's Theorem)**.** *If f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then $\exists c \in (a, b)$ with $f'(c) = 0$.*

Theorem 2.5 (Mean Value Theorem)**.** *If f is continuous on $[a, b]$ and differentiable on (a, b), then $\exists c \in (a, b)$ with*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Theorem 2.6 (Generalized MVT)**.** *If f, g are continuous on $[a, b]$ and differentiable on (a, b), then $\exists c \in (a, b)$ with*

$$[f(b) - f(a)]\, g'(c) = [g(b) - g(a)]\, f'(c).$$

Proof of GMVT. Apply Rolle's to $h(t) = [f(b) - f(a)]g(t) - [g(b) - g(a)]f(t)$. Check: $h(a) = [f(b) - f(a)]g(a) - [g(b) - g(a)]f(a) = f(b)g(a) - f(a)g(b)$, and $h(b)$ gives the same value. So $h(a) = h(b)$, and Rolle's gives $h'(c) = 0$. □

Key Insight

The GMVT is MVT with two functions. It says: the ratio of the increments of f and g equals the ratio of derivatives at some interior point. Setting $g(t) = t$ recovers the ordinary MVT.

2.4 Applications of the MVT

Theorem 2.7 (MVT consequences)**.**

- *$f' = 0$ on $(a, b) \Rightarrow f$ is constant on (a, b).*
- *$f' \geq 0$ on $(a, b) \Rightarrow f$ is nondecreasing.*
- *$|f(b) - f(a)| \leq \sup_{(a,b)} |f'| \cdot |b - a|$ (MVT as a bound).*
- *$f'(a) < \lambda < f'(b) \Rightarrow \exists x \in (a, b)$ with $f'(x) = \lambda$ (**IVT for derivatives**).*

Technique: MVT as a Bounding Tool

Whenever you need to bound $|f(x) - f(y)|$, use the MVT: $|f(x) - f(y)| \leq \sup |f'| \cdot |x - y|$. This turns derivative information into function value bounds. Especially powerful when $f(a) = 0$: then $|f(x)| \leq \sup |f'| \cdot |x - a|$.

2.5 L'Hôpital's Rule

Theorem 2.8 (L'Hôpital)**.** *Suppose $f(a) = g(a) = 0$ and $g'(x) \neq 0$ near a. If $\lim_{x\to a} f'(x)/g'(x) = A$ exists, then $\lim_{x\to a} f(x)/g(x) = A$.*

Proof idea. The GMVT applied to f, g on $[a, x]$ gives $f(x)/g(x) = f'(c)/g'(c)$ for some $c \in (a, x)$. As $x \to a$, $c \to a$ too, and the right side $\to A$. □

Watch Out

L'Hôpital requires the form $0/0$ (or ∞/∞) and requires that $\lim f'/g'$ exists. If the derivative ratio doesn't converge, L'Hôpital says nothing—the original limit may still exist.

2.6 Taylor's Theorem

Theorem 2.9 (Taylor's Theorem with Lagrange Remainder)**.** *Suppose $f^{(n-1)}$ is continuous on $[a, b]$ and $f^{(n)}$ exists on (a, b). Let $\alpha, \beta \in [a, b]$. Define the degree-$(n-1)$ Taylor polynomial*

$$P(t) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!}(t-\alpha)^k.$$

Then there exists x between α and β such that

$$f(\beta) = P(\beta) + \underbrace{\frac{f^{(n)}(x)}{n!}(\beta-\alpha)^n}_{\textit{Lagrange remainder}}.$$

Proof idea. Define M by $f(\beta) = P(\beta) + M(\beta-\alpha)^n$. Let $g(t) = f(t) - P(t) - M(t-\alpha)^n$. Then $g(\alpha) = g(\beta) = 0$ and $g^{(k)}(\alpha) = 0$ for $k = 0, \ldots, n-1$. Apply MVT n times: $g(\alpha) = g(\beta) = 0$ gives $g'(x_1) = 0$, then $g'(\alpha) = g'(x_1) = 0$ gives $g''(x_2) = 0$, and so on. After n steps, $g^{(n)}(x_n) = 0$. But $g^{(n)}(t) = f^{(n)}(t) - M \cdot n!$, so $M = f^{(n)}(x_n)/n!$. □

Key Insight

Taylor's theorem is a quantitative version of "smooth functions are locally polynomial." The Lagrange remainder tells you the error when you truncate the Taylor series. **Key use:** to bound the error in approximating f near α:

$$|f(\beta) - P(\beta)| \leq \frac{\sup|f^{(n)}|}{n!}|\beta-\alpha|^n.$$

As $n \to \infty$, this $\to 0$ if f is well-behaved (entire functions like $e^x, \sin, \cos$).

2.7 Fixed Points and Contraction Mapping

Theorem 2.10. *If $f'(t) \neq 1$ for all t, then f has at most one fixed point ($f(x) = x$).*

Proof. If $f(x) = x$ and $f(y) = y$ with $x \neq y$, MVT gives $f'(c) = (f(y) - f(x))/(y - x) = 1$, contradiction. $\square$

Theorem 2.11 (Contraction Mapping)**.** *If $|f'(t)| \leq A < 1$ for all t, then for any starting point x_1, the iteration $x_{n+1} = f(x_n)$ converges to the unique fixed point of f.*

Key Insight

Why does iteration work? The MVT gives $|x_{n+1} - x_n| = |f(x_n) - f(x_{n-1})| \leq A|x_n - x_{n-1}|$. So the gaps shrink geometrically: $|x_{n+1} - x_n| \leq A^{n-1}|x_2 - x_1|$. The tail sum is bounded by a geometric series, making $\{x_n\}$ Cauchy.

Technique: Gronwall / Bootstrapping

Setup: You have $|f'| \leq A|f|$ and $f(a) = 0$. You want to show $f \equiv 0$.
Step 1 (Local): Let $M_0 = \sup_{[a,x_0]} |f|$. By MVT, $|f(x)| \leq A|f(c)||x - a| \leq AM_0(x_0 - a)$. Taking sup: $M_0 \leq AM_0(x_0 - a)$. If $x_0 - a < 1/A$, then $AM_0(x_0 - a) < M_0$, forcing $M_0 = 0$. So $f \equiv 0$ on $[a, x_0]$.
Step 2 (Global): Restart at x_0 as the new base point (we now know $f(x_0) = 0$). Cover $[a, b]$ in finitely many steps of length $< 1/A$. Conclude $f \equiv 0$ on all of $[a, b]$.
Corollary: If f and g both satisfy $h' = Ah$ and $h(a) = 0$, apply Gronwall to $h = f - g$ to get $f = g$ (uniqueness of ODE solutions).

3 The Riemann Integral

Big Picture

The Riemann integral is defined by approximating the area under f from above (upper sums) and below (lower sums). If the approximations can be made to agree, f is integrable. The key tool is the **integrability criterion**: $f \in \mathscr{R}$ iff you can make $U - L < \varepsilon$ for any $\varepsilon > 0$.

3.1 Partitions and Darboux Sums

Definition 3.1. A **partition** of $[a, b]$ is $P = \{x_0, x_1, \ldots, x_n\}$ with $a = x_0 \leq \cdots \leq x_n = b$. Let $\Delta x_i = x_i - x_{i-1}$, $M_i = \sup_{[x_{i-1},x_i]} f$, $m_i = \inf_{[x_{i-1},x_i]} f$.

The **upper** and **lower Darboux sums** are:

$$U(P, f) = \sum_{i=1}^{n} M_i \Delta x_i, \qquad L(P, f) = \sum_{i=1}^{n} m_i \Delta x_i.$$

Always $L(P, f) \leq U(P, f)$, and refining P (adding more points) can only decrease U and increase L.

Definition 3.2. The **upper Riemann integral** is $\overline{\int} f = \inf_P U(P, f)$ and the **lower Riemann integral** is $\underline{\int} f = \sup_P L(P, f)$.

f is **Riemann integrable** ($f \in \mathscr{R}$) iff $\overline{\int} f = \underline{\int} f$.

3.2 The Integrability Criterion

Theorem 3.3 (Integrability Criterion). *$f \in \mathscr{R}$ if and only if for every $\varepsilon > 0$, there exists a partition P such that*

$$U(P, f) - L(P, f) < \varepsilon.$$

Key Insight

$U - L = \sum(M_i - m_i)\Delta x_i$ is the **total oscillation** of f. The criterion says f is integrable iff you can make the total oscillation arbitrarily small by choosing a fine enough partition. Oscillation is large where f is discontinuous, so roughly: f is integrable iff it's not too discontinuous.

Technique: Proving $f \in \mathscr{R}$ via the Criterion

To show $f \in \mathscr{R}$, construct a partition P with $U(P, f) - L(P, f) < \varepsilon$. The standard strategy:

1. Split intervals into "bad" (where f oscillates a lot) and "good" (where f is nearly constant).
2. On bad intervals: the oscillation can be large, but make the total *length* small (say $< \delta$), contributing $\leq 2M\delta$ to $U - L$.
3. On good intervals: oscillation $\leq \varepsilon$ each, contributing $\leq \varepsilon(b - a)$ total.
4. Choose $\delta = \varepsilon/(2M)$ so both contributions are $O(\varepsilon)$.

3.3 Who Is Integrable?

Theorem 3.4. *The following classes of functions are in $\mathscr{R}([a, b])$:*

- ***Continuous functions*** *$f \in C([a, b])$: continuity on a compact set is uniform, so you can make oscillation on each interval $< \varepsilon$ simultaneously.*
- ***Monotone functions****: at most countably many discontinuities, all of jump type.*
- ***Bounded functions whose discontinuities form a zero set****.*

The Dirichlet function ($f = 1$ on $\mathbb{Q}$, $f = 0$ on $\mathbb{R} \setminus \mathbb{Q}$) is not *integrable: $U = 1, L = 0$ for every partition.*

Definition 3.5. A set $E \subset \mathbb{R}$ is a **zero set** (measure zero) if for every $\varepsilon > 0$, there exist countably many intervals $\{I_n\}$ covering E with $\sum |I_n| < \varepsilon$.

Example 3.6. $\mathbb{Q}$ is a zero set (it's countable: enumerate and cover the n-th rational by an interval of length $\varepsilon/2^n$). The Thomae function ($f(p/q) = 1/q$, $f(\text{irrational}) = 0$) is integrable: its discontinuities are exactly $\mathbb{Q}$, a zero set.

3.4 Riemann–Stieltjes Integration

Big Picture

The Riemann–Stieltjes integral $\int f\,d\alpha$ generalizes the Riemann integral by replacing the length measure dx with a measure $d\alpha$ induced by a monotone increasing function α. If α has a derivative, the Stieltjes integral reduces to an ordinary Riemann integral.

Replace $\Delta x_i = x_i - x_{i-1}$ by $\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1})$ to get $U(P, f, \alpha)$ and $L(P, f, \alpha)$. The integrability criterion and classes remain the same.

Theorem 3.7 (Reduction to Riemann)**.** *If $\alpha' \in \mathscr{R}$ and f is bounded, then $f \in \mathscr{R}(\alpha)$ iff $f\alpha' \in \mathscr{R}$, and*

$$\int_a^b f\,d\alpha = \int_a^b f\alpha'\,dx.$$

Proof idea. By the MVT, $\Delta\alpha_i = \alpha'(t_i)\Delta x_i$ for some $t_i \in [x_{i-1}, x_i]$. So a Stieltjes sum $\sum f(s_i)\Delta\alpha_i = \sum f(s_i)\alpha'(t_i)\Delta x_i$ is close to the Riemann sum $\sum f(s_i)\alpha'(s_i)\Delta x_i$ for $f\alpha'$. The error is bounded by $M[U(P, \alpha') - L(P, \alpha')]$, which can be made small since $\alpha' \in \mathscr{R}$. □

Theorem 3.8 (Integration by Parts)**.**

$$\int_a^b f\,d\alpha + \int_a^b \alpha\,df = f(b)\alpha(b) - f(a)\alpha(a).$$

Theorem 3.9 (Change of Variable)**.** *Let $\varphi : [A, B] \to [a, b]$ be strictly increasing. Let $\beta = \alpha \circ \varphi$ and $g = f \circ \varphi$. Then*

$$\int_A^B g\,d\beta = \int_a^b f\,d\alpha.$$

3.5 Improper Integrals

Definition 3.10. $\displaystyle\int_a^\infty f\,dx = \lim_{R\to\infty} \int_a^R f\,dx$, if the limit exists. Similarly for $\int_{-\infty}^b$ and for singularities at endpoints.

Theorem 3.11 (Comparison)**.** *If $0 \le f(x) \le g(x)$ and $\int g$ converges, then $\int f$ converges.*

Key Insight

Key benchmark integrals:

- $\displaystyle\int_1^\infty \frac{dx}{x^p}$ converges iff $p > 1$.
- $\displaystyle\int_0^1 \frac{dx}{x^p}$ converges iff $p < 1$.

The switchover happens at $p = 1$ (the harmonic integral). Near ∞, you need f to decay faster than $1/x$; near 0, you need f to blow up slower than $1/x$.

4 Hölder's Inequality and the L^2 Space

Big Picture

Hölder's inequality is the integral version of the AM-GM inequality. It lets you bound $\int fg$ in terms of norms of f and g separately. The special case $p = q = 2$ is Cauchy–Schwarz. Hölder is the key to proving the triangle inequality for the L^2 norm.

4.1 Young's Inequality and Hölder

Theorem 4.1 (Young's Inequality)**.** *For $a, b \geq 0$ and conjugate exponents $1/p + 1/q = 1$ with $p, q > 1$:*

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Proof. Fix b and optimize $ab - a^p/p$ over $a \geq 0$. The maximum is at $a = b^{1/(p-1)} = b^{q-1}$, giving $b^q - b^q/p = b^q/q$. Alternatively: Young is equivalent to ln being concave, via $\ln(ab) = \ln a + \ln b \leq p \ln a/p + q \ln b/q = \ln(a^p/p + b^q/q)$ by Jensen. □

Theorem 4.2 (Hölder's Inequality)**.** *Let $1/p + 1/q = 1$ with $p, q > 1$. If $f \in \mathscr{R}(\alpha)$ and $g \in \mathscr{R}(\alpha)$, then*

$$\int_a^b |fg| \, d\alpha \leq \left(\int_a^b |f|^p \, d\alpha \right)^{1/p} \left(\int_a^b |g|^q \, d\alpha \right)^{1/q} = \|f\|_p \|g\|_q.$$

Proof. If $\|f\|_p = 0$ or $\|g\|_q = 0$, both sides are 0. Otherwise, normalize: let $u = |f|/\|f\|_p$ and $v = |g|/\|g\|_q$. Apply Young pointwise: $u(x)v(x) \leq u(x)^p/p + v(x)^q/q$. Integrate: $\int uv \leq 1/p + 1/q = 1$. Multiply back: $\int |fg| \leq \|f\|_p \|g\|_q$. □

Key Insight

The key steps to remember:

1. Young's inequality: $uv \leq u^p/p + v^q/q$ (from concavity of ln).
2. Normalize so that $\|u\|_p = \|v\|_q = 1$.
3. After normalization, Young + integration gives $\int uv \leq 1$.
4. Denormalize (multiply back by $\|f\|_p \|g\|_q$).

4.2 The L^2 Space and Its Triangle Inequality

Definition 4.3. For $f \in \mathscr{R}(\alpha)$, the L^2 **norm** (with respect to α) is

$$\|f\|_2 = \left(\int_a^b |f|^2 \, d\alpha \right)^{1/2}.$$

Theorem 4.4 (Triangle Inequality in L^2)**.** $\|f+g\|_2 \le \|f\|_2 + \|g\|_2$.

Proof. Square both sides. Need to show $\|f+g\|_2^2 \le (\|f\|_2 + \|g\|_2)^2$. Expand the left side:

$$\|f+g\|_2^2 = \int |f+g|^2 \, d\alpha = \int f^2 + 2\int fg + \int g^2.$$

Expand the right side: $(\|f\|_2 + \|g\|_2)^2 = \|f\|_2^2 + 2\|f\|_2\|g\|_2 + \|g\|_2^2$.

Everything cancels except the cross terms: need $\int fg \le \|f\|_2\|g\|_2$. This is Cauchy–Schwarz (Hölder with $p = q = 2$). Take the square root. □

Technique: Triangle Inequality via Squaring

Template: to prove $\|A\| \le \|B\| + \|C\|$, square both sides, expand, bound the cross term by Schwarz/Hölder, recognize a perfect square, take the root. This pattern appears whenever you have an inner product or L^2 norm.

4.3 L^2 Approximation by Piecewise Linear Functions

Theorem 4.5 (Approximation in L^2)**.** *If $f \in \mathscr{R}(\alpha)$ on $[a,b]$ and $\varepsilon > 0$, there exists a continuous function g on $[a,b]$ with $\|f-g\|_2 < \varepsilon$.*

Proof. **Construction:** Since $f \in \mathscr{R}(\alpha)$, choose P with $U(P,f,\alpha) - L(P,f,\alpha) < \varepsilon^2/(2M)$ where $M = \sup|f|$. Define g by linear interpolation between the values of f at the partition points; g is continuous.

Bound: On each $[x_{i-1}, x_i]$, both f and g (being a convex combination of f-values at partition points) lie in $[m_i, M_i]$. So $|f(t) - g(t)| \le M_i - m_i$ pointwise. Then:

$$\begin{aligned}
\|f-g\|_2^2 &= \int |f-g|^2 \, d\alpha \\
&\le \sum_{i=1}^{n} (M_i - m_i)^2 \, \Delta\alpha_i \\
&\le 2M \sum_{i=1}^{n} (M_i - m_i) \, \Delta\alpha_i \qquad (\text{since } M_i - m_i \le 2M) \\
&= 2M[U(P,f,\alpha) - L(P,f,\alpha)] \\
&< 2M \cdot \frac{\varepsilon^2}{2M} = \varepsilon^2.
\end{aligned}$$

Take the square root: $\|f-g\|_2 < \varepsilon$. □

Key Insight

The key algebraic trick: $(M_i - m_i)^2 \le 2M(M_i - m_i)$. One factor of oscillation is bounded by the global constant $2M$, converting squared oscillation to linear oscillation so the Riemann criterion applies.

Why piecewise linear? Linear interpolation is the simplest way to build a continuous function that passes through prescribed values at the partition points, and it automatically stays within $[m_i, M_i]$ on each interval (by convexity).

Technique: Applying Hölder: How to Split h

Given $\int h$, write $h = f \cdot g$ and choose p, q so that both $\|f\|_p$ and $\|g\|_q$ are computable/finite. Checklist:

- Identify what power of h appears naturally.
- Set $|f|^p =$ (something you can integrate) and $|g|^q =$ (something else).
- Verify $1/p + 1/q = 1$.
- Apply Hölder.

Example: To show $\left(\int fg\right)^2 \leq \int f^2 \cdot \int g^2$, write $fg = f \cdot g$ and apply Hölder with $p = q = 2$.

5 Distance Functions and Metric Continuity

Big Picture

Distance to a set $\rho_E(x) = \inf_{e \in E} d(x, e)$ is one of the most useful constructions in metric topology. It creates continuous functions from closed sets, and it lets you separate disjoint closed sets smoothly (Urysohn's lemma).

5.1 Distance to a Set

Definition 5.1. For a nonempty set E in a metric space (X, d), the **distance function** is

$$\rho_E(x) = \inf_{e \in E} d(x, e).$$

Theorem 5.2 (Properties of ρ_E). *1. $\rho_E(x) = 0 \iff x \in \overline{E}$ (the closure of E).*

*2. $|\rho_E(x) - \rho_E(y)| \leq d(x, y)$ (**Lipschitz-1**, hence uniformly continuous).*

Proof of (2). For any $e \in E$: $d(x, e) \leq d(x, y) + d(y, e)$. Taking the inf over e: $\rho_E(x) \leq d(x, y) + \rho_E(y)$. So $\rho_E(x) - \rho_E(y) \leq d(x, y)$. Swap x and y for the other direction. $\square$

Key Insight

The proof uses a fundamental trick: **triangle inequality + inf**. The inf over a third variable preserves the triangle inequality. This pattern appears in many metric space arguments.

5.2 Urysohn Separation Function

Theorem 5.3 (Urysohn-type)**.** *Let A and B be disjoint closed sets in a metric space. Define*

$$f(p) = \frac{\rho_A(p)}{\rho_A(p) + \rho_B(p)}.$$

Then f is continuous, $f|_A = 0$, and $f|_B = 1$.

Proof. The denominator $\rho_A(p) + \rho_B(p) > 0$ for all p: if it were 0, then $\rho_A(p) = \rho_B(p) = 0$, so $p \in \overline{A} \cap \overline{B} = A \cap B = \emptyset$, contradiction. Since ρ_A and ρ_B are continuous (Lipschitz-1) and the denominator is positive and continuous, f is continuous. On A: $\rho_A = 0$, so $f = 0$. On B: $\rho_B = 0$, so $f = 1$. □

Key Insight

This is remarkable: from a purely topological fact (two closed sets are disjoint) you construct a continuous function that perfectly separates them. The function f takes values in $[0, 1]$, is 0 on one set, 1 on the other, and varies continuously between them.

6 Master Technique Reference

This section is a catalog of the core proof patterns. For each exam problem, identify which pattern applies first.

Technique: Pattern 1: Showing a Series Converges

1. **Root test:** compute $\limsup |a_n|^{1/n}$. Works for n-th powers.
2. **Ratio test:** compute $\limsup |a_{n+1}/a_n|$. Works for factorials.
3. **Comparison:** bound $|a_n| \leq c_n$ where $\sum c_n$ is known.
4. **Cauchy condensation:** for decreasing $a_n \geq 0$, replace by $\sum 2^k a_{2^k}$.
5. **p-series benchmark:** $\sum 1/n^s$ converges iff $s > 1$.

Technique: Pattern 2: Showing $f \in \mathscr{R}$

1. **Continuous:** $f \in C([a, b]) \Rightarrow f \in \mathscr{R}$.
2. **Monotone:** monotone $\Rightarrow f \in \mathscr{R}$.
3. **Integrability criterion:** exhibit P with $U(P, f) - L(P, f) < \varepsilon$. Strategy: isolate bad intervals (large oscillation) into small total length; bound good intervals by uniform small oscillation.
4. **Zero set:** if discontinuities of bounded f form a zero set, then $f \in \mathscr{R}$.
5. **Sandwich:** find $g \leq f \leq h$ with $g, h \in \mathscr{R}$ and $\int (h - g) < \varepsilon$.

Technique: Pattern 3: Bounding with MVT

Given: f differentiable, $|f'| \leq M$. Then: $|f(x) - f(y)| \leq M|x - y|$ for all x, y. Special case: $f(a) = 0 \Rightarrow |f(x)| \leq M|x-a|$. Use this: to bound function values, to show Lipschitz continuity, to prove convergence in iteration arguments.

Technique: Pattern 4: Uniqueness via Subtraction

Want: $f = g$ given both satisfy some conditions. Method: Let $h = f - g$. Show h satisfies conditions implying $h \equiv 0$ (e.g., Gronwall: $h(a) = 0$ and $|h'| \leq A|h|$).

Technique: Pattern 5: Applying Hölder

Want to bound $\int |fg|$ or $\int |h|$ (by writing $h = f \cdot g$). Steps: (1) pick p and $q = p/(p-1)$ with $1/p+1/q = 1$; (2) write $h = f \cdot g$ with $|f|^p$ and $|g|^q$ both integrable; (3) apply $\int |fg| \leq \|f\|_p \|g\|_q$. Most common: $p = q = 2$ (Cauchy–Schwarz).

Technique: Pattern 6: Constructing the $U - L$ Bound for R-S

When α has a derivative: use $\Delta\alpha_i = \alpha'(t_i)\Delta x_i$ (MVT). Compare R-S sums to Riemann sums for $f\alpha'$; the error is $\leq M[U(P, \alpha') - L(P, \alpha')]$, which $\to 0$ since $\alpha' \in \mathscr{R}$.

7 Common Pitfalls and Quick Tests

Watch Out

Pitfall 1: Converse of necessary conditions.

- $a_n \to 0$ does NOT imply $\sum a_n$ converges (harmonic series).
- f continuous does NOT imply f differentiable ($|x|$ at 0).
- $f \in \mathscr{R}$ does NOT imply f is continuous (step functions, Thomae).

Watch Out

Pitfall 2: Root/ratio test gives $L = 1$. The test is *inconclusive*, not "converges." You need a different test. Both $\sum 1/n$ (diverges) and $\sum 1/n^2$ (converges) give $L = 1$ for the root test.

Watch Out

Pitfall 3: Forgetting to check endpoints for power series. $\sum c_n(x - a)^n$ converges absolutely for $|x - a| < R$ and diverges for $|x - a| > R$. At the two endpoints $x = a \pm R$, the root/ratio test is inconclusive and you must check directly.

Watch Out

Pitfall 4: Swapping limit and integral. In general, $\lim \int f_n \neq \int \lim f_n$. You need uniform convergence (or dominated convergence) to justify swapping. Watch for this in Stieltjes

reduction arguments.

Watch Out

Pitfall 5: Dirichlet vs. Thomae.

- Dirichlet: $f = 1_{\mathbb{Q}}$. NOT integrable ($U = 1, L = 0$ always).
- Thomae: $f(p/q) = 1/q$, $f(\text{irrational}) = 0$. IS integrable (discontinuities are $\mathbb{Q}$, a zero set; integral $= 0$).

7.1 Quick Reference Card

Series tests ranked by power: Root test $\geq$ Ratio test $>$ Comparison $>$ n-th term test.

p-series benchmarks: $\sum 1/n^s$ conv. iff $s > 1$. $\int_1^\infty x^{-p}$ conv. iff $p > 1$. $\int_0^1 x^{-p}$ conv. iff $p < 1$.

Integrability hierarchy: $C([a, b]) \subset \mathscr{R} \supset \{\text{monotone}\} \supset \{\text{zero-set discontinuities}\}$.

Key inequalities (all follow from AM-GM / convexity):

$$\text{AM-GM} \Rightarrow \text{Young } (uv \leq u^p/p + v^q/q) \Rightarrow \text{Hölder} \Rightarrow \text{Cauchy–Schwarz}.$$

MVT family: Rolle's $\Rightarrow$ MVT $\Rightarrow$ GMVT $\Rightarrow$ L'Hôpital. All follow by applying Rolle's to a cleverly chosen auxiliary function.